Filters

Store

WI_1

Category

HOUSEHOLD

What-if scenario

Price change (%)

-16

Assumed price elasticity

-1.30

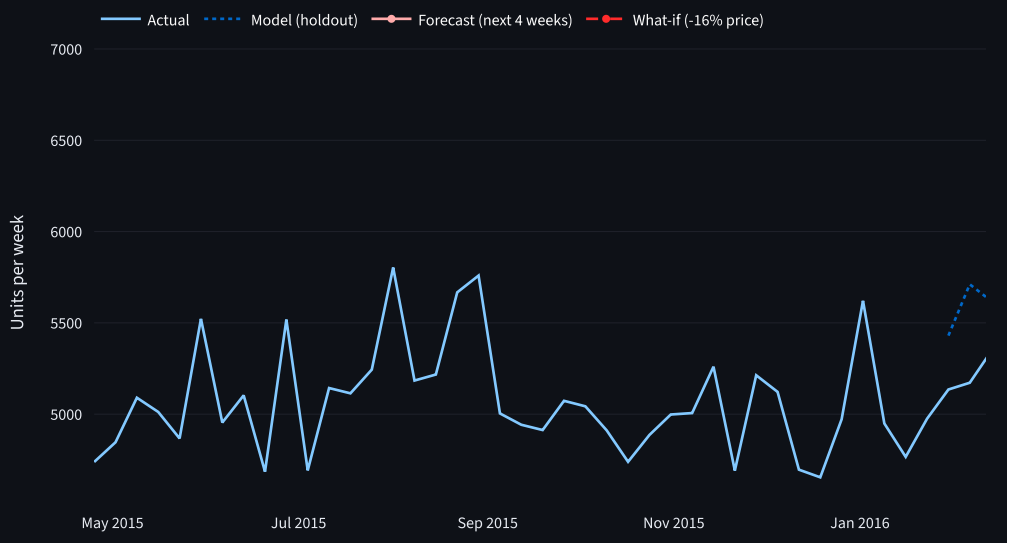
Service level (z-score)

1.75

Retail Demand Forecasting & Inventory Optimization

M5 Walmart data | LightGBM weekly forecasts | 30-day (4-week) horizon

Last 4 weeks (actual units)	Next 4 weeks (forecast)	Next 4 weeks (what-if)	Holdout WMAPE (12 weeks)
21,541	21,221	26,620	5.3%
	↓ -1.5% vs last 4 weeks	↑ +25.4% vs baseline	



Inventory recommendation (next 4 weeks)

store_id	cat_id	dept_id	forecast_units	scenario_units	safety_stock
WI_1	HOUSEHOLD	HOUSEHOLD_1	16679	20922	
WI_1	HOUSEHOLD	HOUSEHOLD_2	4542	5697	

Recommended stock = what-if forecast for 4 weeks + safety stock (service-level z x holdout error std x sqrt(4 weeks)). Price effect uses the elasticity assumption in the sidebar, not an estimated value.

Download recommendation CSV